Adeem Haris

+91-7838284514 | adeemharis@gmail.com | adeemharis
algorist_adeem | MyWebsite | adeemharis

Education

- **Indian Institute of Technology Delhi** *July 2026 - Present*
  *M.Tech - **Cybersecurity*** Delhi, India
  - Learning how to secure modern AI, Networking and Computing Systems
- **Indian Institute of Technology Jodhpur** *Dec 2021 - May 2025*
  *B.Tech - **Artificial Intelligence and Data Science*** Jodhpur, India
  - Minor in Mathematical and Computaional Economics

Experience

- **HCLTech** *Sep 2025 - August 2026*
  Technical Lead - GenAI Hyderabad, India
  - Led GenAI engineering initiatives for a Fortune 500 fintech client (**Western Union**), designing and shipping production-grade **AI automation systems across DevOps, HR and enterprise workflow** domains
  - Built an **AI-powered code review bot** integrating LLM-based static analysis into the CI/CD pipeline, **reducing manual PR review time** and improving defect detection pre-merge
  - Completed GenAI Training with coverage of RAGs, Prompt Engineering, Agentic AI and Foundational Models
  - Designed a **scoped AI agent authentication system** for enterprise resource access, implementing short-lived, permission-bound credentials (OAuth2/PAM-based) to enable **safe autonomous agent operations within corporate infrastructure**
  - Developed a **full-stack Attendance Compliance Automation platform** with role-based dashboards for employees and managers, automated leave/WFH approval workflows, and real-time compliance tracking — **replacing a fully manual process**
- **Weya.AI** *May 2024 - July 2024*
  Machine Learning Intern Remote
  - Designed and implemented a complete **end-to-end machine learning pipeline for energy disaggregation**, covering data collection, preprocessing, feature engineering, model training, and cloud deployment for real-world client usage.
  - Conducted research on advanced generative models (e.g., **GANs, VAEs**) and SOTA **NILM** (Non-Intrusive Load Monitoring) approaches, optimizing architectures to achieve client-specific accuracy targets .
  - Worked on defining data requirements, improving data quality, and ensuring seamless integration between ML models and developed dataset.

Skills

- **Programming Languages:** C++, Java, Python, Javascript, Go, R
- **Web Technologies:** Node.js, React.js, Express.js, Django, Angular, Flask, FastAPI, GraphQL
- **Database Systems:** MySql, MongoDB, PostgreSQL, Elasticsearch, Redis, Neo4j
- **Data Science & Machine Learning:** Pytorch, Scikit-Learn, Tensorflow, HuggingFace, Pandas, Numpy
- **Cloud Technologies:** AWS, Google Cloud Platform, Firebase, Docker, Vercel
- **DevOps & Version Control:** Git, MLflow, CI/CD Pipeline, DevOps, MlOps
- **Specialized Area:** High Performance Computing, CUDA, Parallel Processing
- **Mathematical & Statistical Tools:** RStudio, Matlab

Scores and Achievements

- **GATE Computer Science and Information Technology 2024** *Mar 2024*
  - Secured an All India Rank of **468** among 123,967 candidates [🌐]
- **GATE Data Science and Artificial Intelligence 2024** *Mar 2024*
  - Secured an All India Rank of **168** among 59,210 candidates [🌐]
- **JEE Advanced 2021** *Nov 2021*
  - Secured an All India Rank of **3716** among a million candidates [🌐]
- **KVPY SA Fellow 2019** *Nov 2021*
  conducted by IISc Bangalore [🌐]
  - Secured an All India Rank of **864**

Projects

- **Accelerated Deep Learning**
  Tools: C++, CUDA, GPU Programming, High Performance Computing
  - Optimized deep learning inference pipeline in C++ using CUDA, reducing computational bottlenecks (70%) through GPU acceleration and parallelization.
  - Worked with **Prof Dip Sankar Banerjee** to accelerate **Frugally-Deep**, a C++ header Library which is used for forward pass in Deep Learning Models trained with Keras .
  - Benchmarked the Frugally-Deep Library to identify the bottleneck functions , taking almost 70% of the computational time for which GPU Programming can be used to accelerate those functions .
- **Video Streaming Platform**
  Tools: Node.js, MongoDB, Full Stack Development, JavaScript, Neo4j, Team Work
  - Developed a full-featured video streaming platform using the **MERN stack (MongoDB, Express.js, React.js, Node.js)** as part of a 3-member team under **Prof Saptarshi Pyne**.
  - Designed and implemented **database models in MongoDB** for storing video metadata, user profiles, playlists, comments, and watch history, ensuring optimized query performance.
  - Integrated **Neo4j graph database for advanced recommendation features** based on user viewing patterns, leveraging graph traversal algorithms for personalized suggestions.
  - Built **RESTful APIs in Node.js and Express.js** for video upload, streaming, authentication, and search functionality, with proper error handling and data validation.
- **RideMate Application Development**
  Tools: Flutter, Firebase, Dart, openstreetmaps API
  - Developed two applications, **User** and **Driver**, for a college-based startup, facilitating ride-sharing services.
  - Implemented **authentication using Firebase Auth** for both user and driver accounts, including phone and email verification.
  - Designed an intuitive UI for users to select pickup and destination locations with a real-time map view powered by the **openstreetsmap API**.
  - **Added ride-sharing functionality** with features like ride confirmation dialogs, shareable ride options, and notifications upon driver assignment.
- **Silicosis Detection from Chest X-rays**
  Tools: Python, PyTorch, ResNet-101, Deep Learning, Transfer Learning
  - Collaborated with **Prof. Richa Singh** to develop a **deep learning model for detecting 29 distinct lung anomalies**, including silicosis, tuberculosis, and other conditions, from chest X-ray images.
  - Used **transfer learning with a pretrained ResNet-101** architecture to achieve accurate multilabel, multiclass classification of chest X-ray images.
  - Processed and annotated the VinDrCXR dataset, applying extensive data preprocessing and augmentation to improve model robustness.
  - Evaluated model performance using metrics such as accuracy, precision, recall, and F1-score to ensure reliable predictions across all anomaly classes.